

CYCLIC SORT

It is used to sort arrays containing range from 1 to N. (0 to N-1)

Algorithm :-

0 1 2 3 4
3, 5, 2, 1, 4

After sorting :- 0 1 2 3 4
1, 2, 3, 4, 5

$\therefore \text{index} = \text{value} - 1$

$\Rightarrow$ 0 1 2 3 4
(3), 5, (2), 1, 4

swap with index 2

swap in

(2) (5) 3 1 4

swap with index 1

(5) 2 3 1 (4)

swap with index 4

(4) 2 3 (1) 5

swap with index 3

1 2 3 4 5

Time Complexity - $O(N)$

Space Complexity - $O(1)$